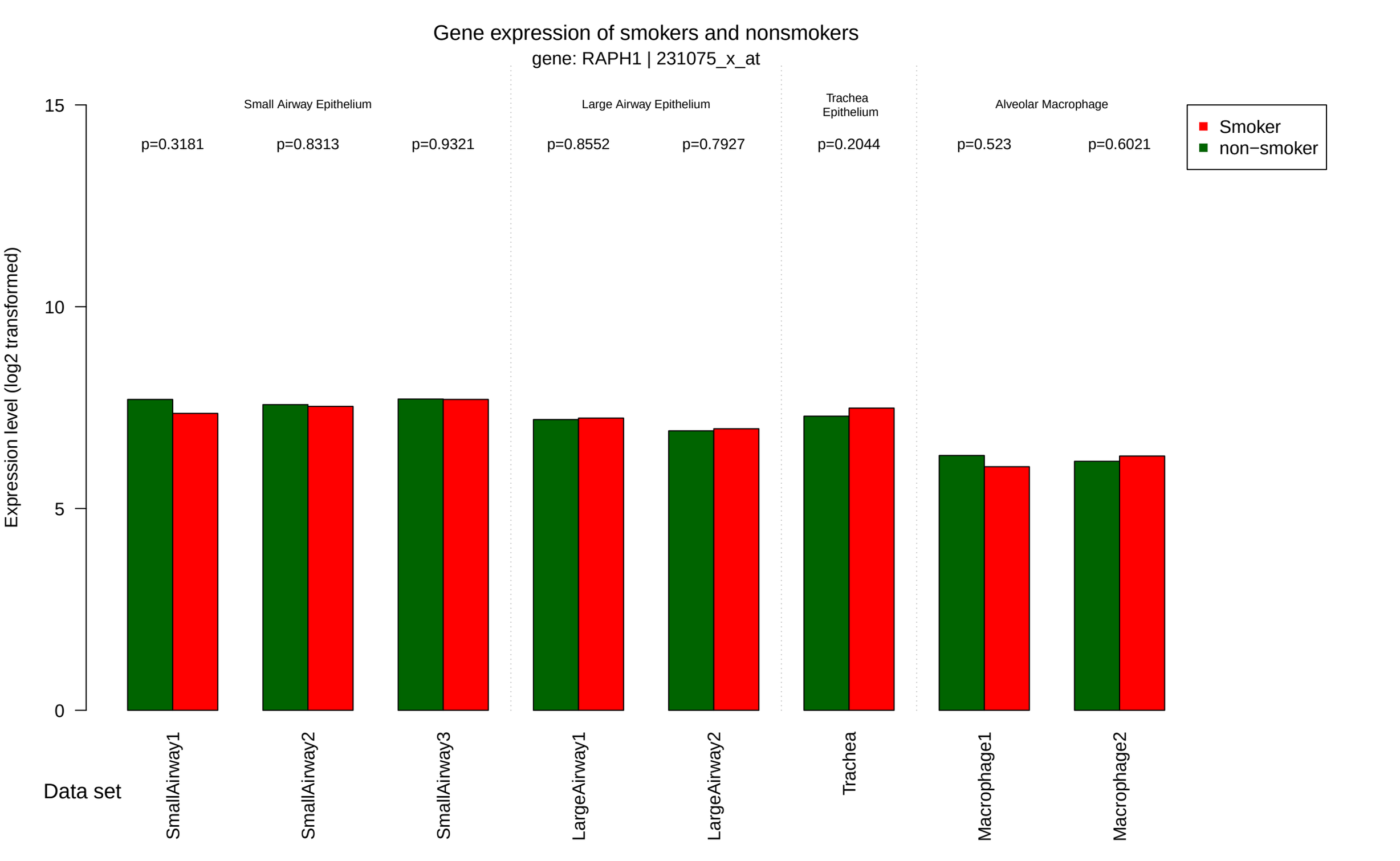
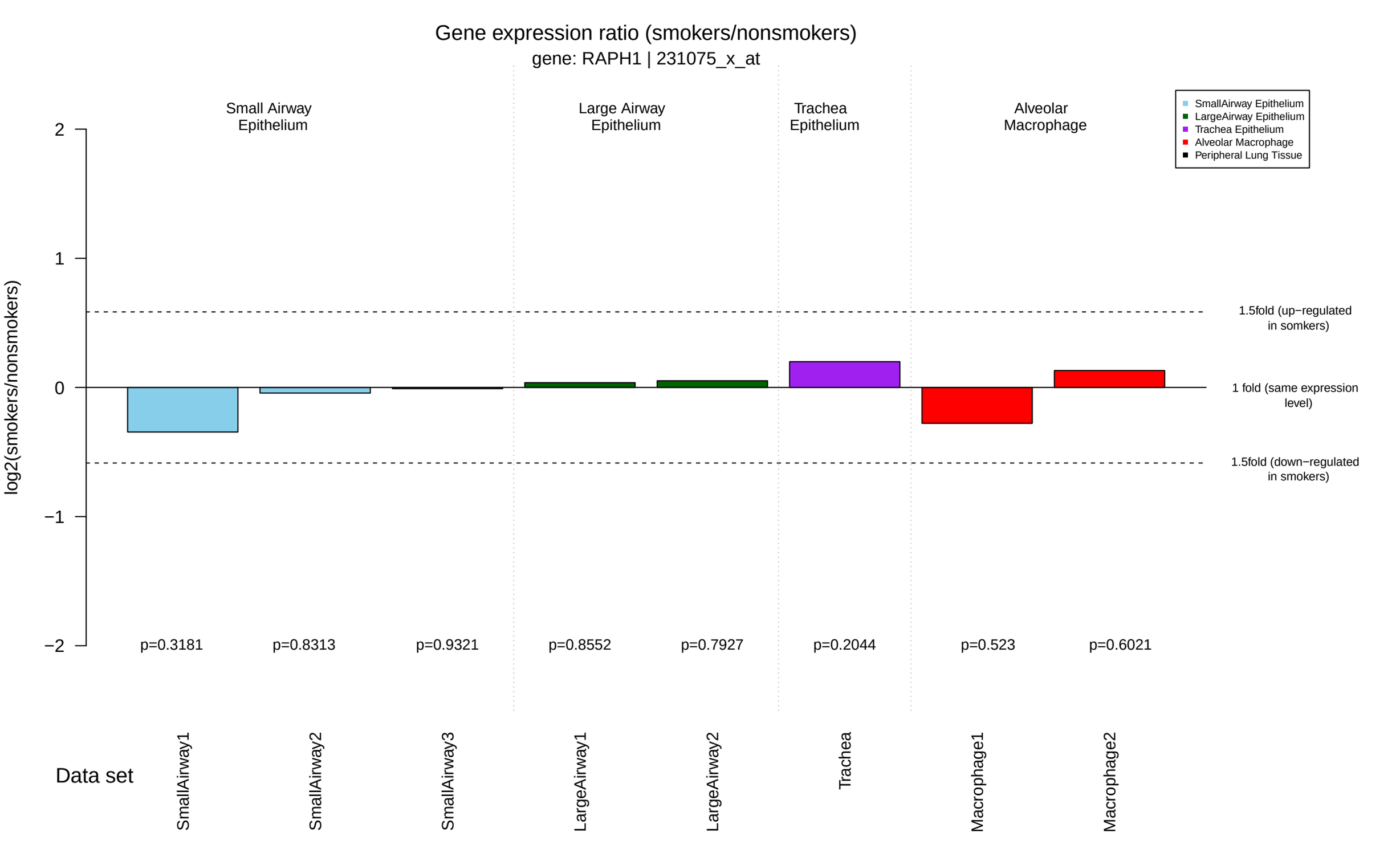
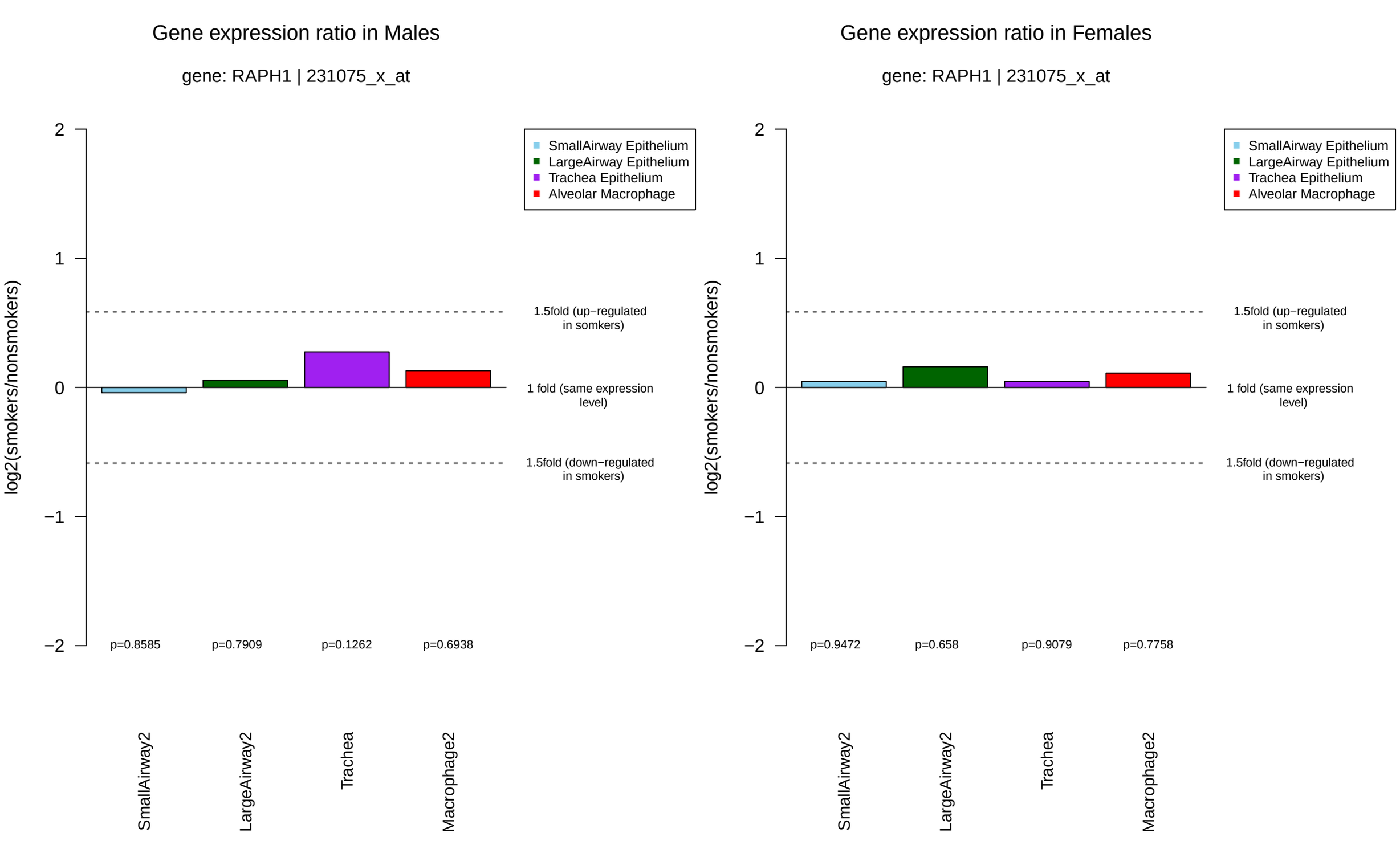
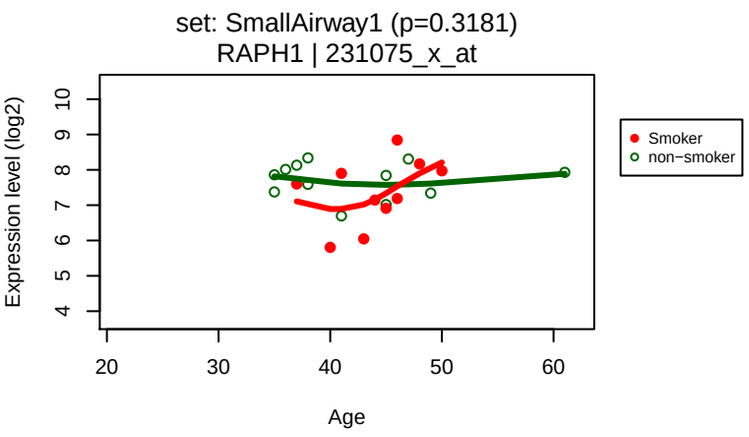
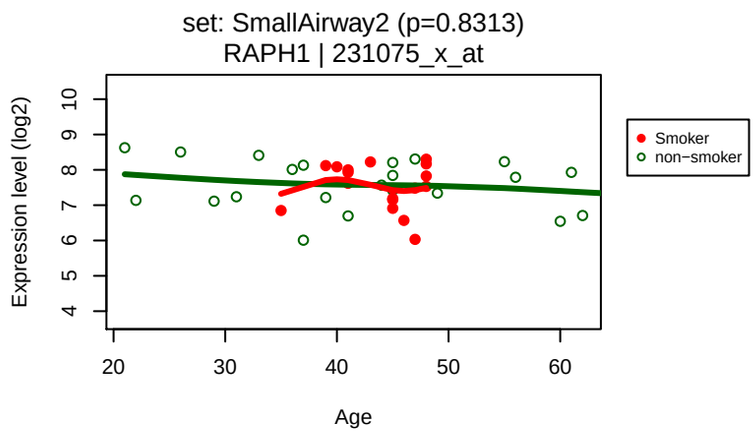
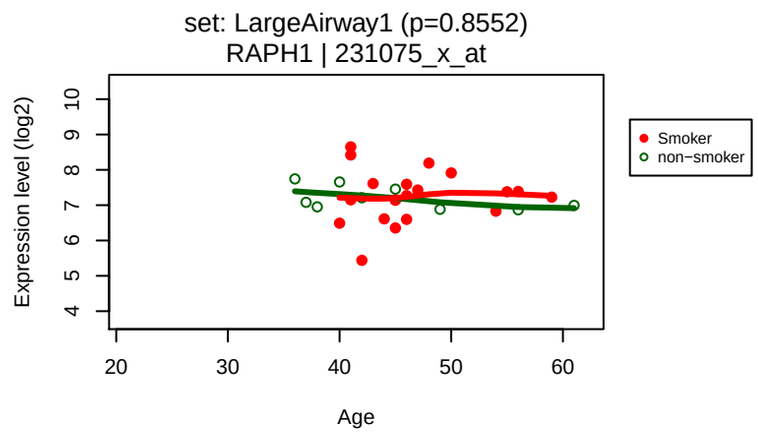

Gene expression of smokers and nonsmokers
gene: RAPH1 | 231075_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3181
p=0.8313
p=0.9321
p=0.8552
p=0.7927
p=0.2044
p=0.523
p=0.6021
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

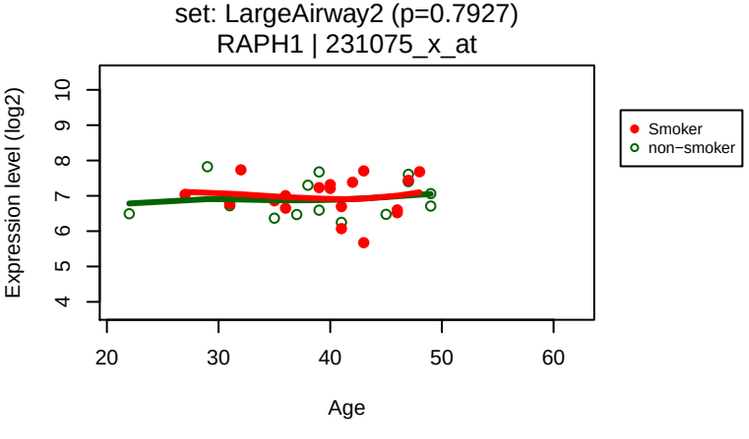

Gene expression ratio (smokers/nonsmokers)
gene: RAPH1 | 231075_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3181
p=0.8313
p=0.9321
p=0.8552
p=0.7927
p=0.2044
p=0.523
p=0.6021
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

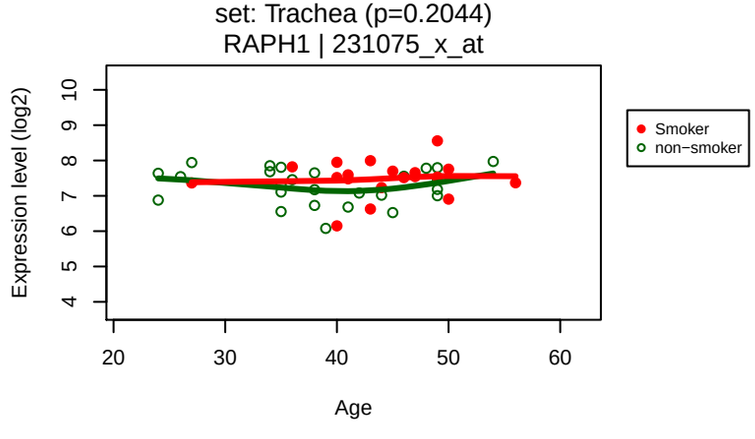

Gene expression ratio in Males
gene: RAPH1 | 231075_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8585
p=0.7909
p=0.1262
p=0.6938
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPH1 | 231075_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9472
p=0.658
p=0.9079
p=0.7758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

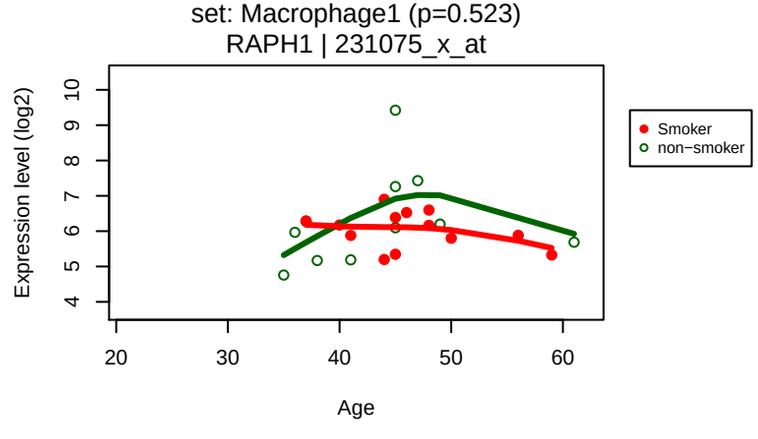

set: SmallAirway1 (p=0.3181)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker

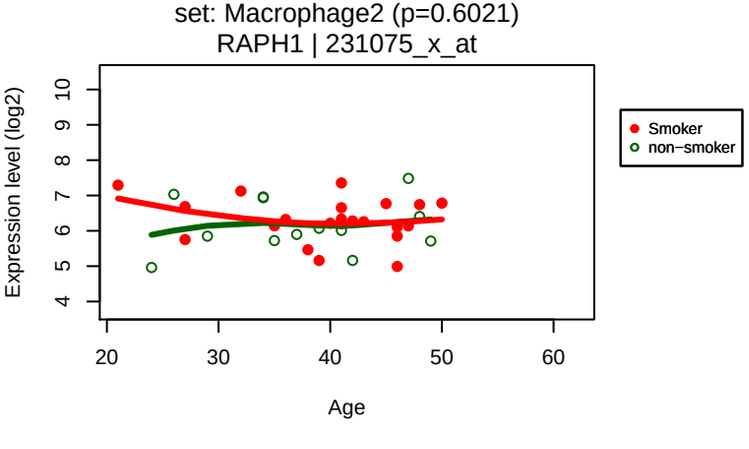

set: SmallAirway2 (p=0.8313)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker

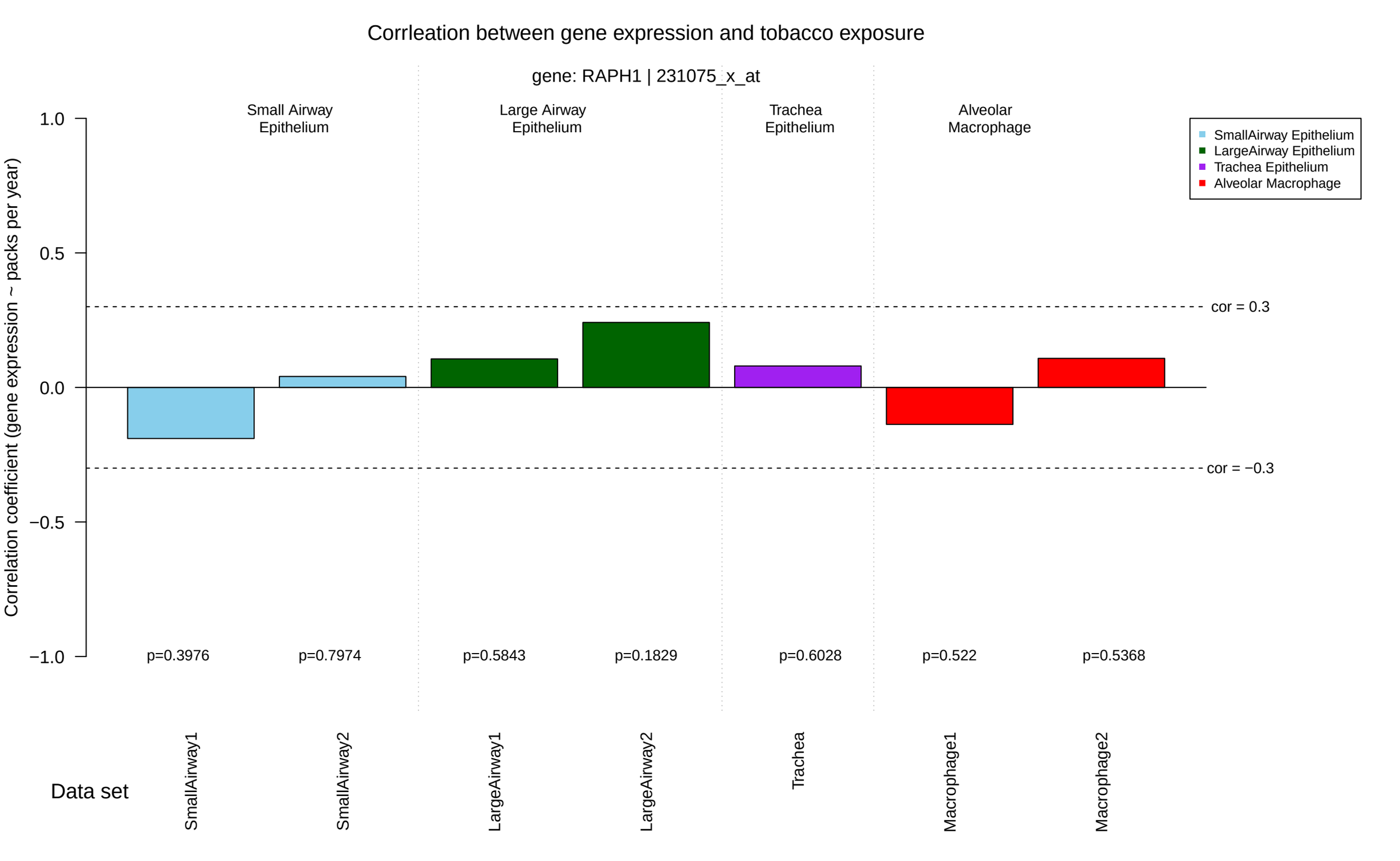

set: LargeAirway1 (p=0.8552)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7927)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2044)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.523)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6021)
RAPH1 | 231075_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPH1 | 231075_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3976
p=0.7974
p=0.5843
p=0.1829
p=0.6028
p=0.522
p=0.5368
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2